AZ-305 · Extras — Missing Topic Locks

Overflow facts · color = source domain: D1 blue · D2 green · D3 orange · D4 purple

D4 VM family decoder

Letter	Meaning	Trigger words
A	Ancient/basic	Cheap dev/test; **no AN**
B	Burstable	Intermittent CPU
D	Default	General-purpose
E/M	Extra/massive memory	DB, cache, analytics / SAP HANA
F	Fast CPU	Compute optimized
L	Local disk	Storage IOPS / local NVMe
N	NVIDIA GPU	NC compute · NV viz · ND deep learning
H	HPC	MPI, CFD, weather, FEA, InfiniBand
DC	Data confidential	TEE / secure enclave

Suffix: s=Premium storage · **d**=local temp disk · **_v#**=generation.

D4 disk SKU by IOPS

Need	Pick
≤500 IOPS, cheapest	**Standard HDD**
≤6K IOPS (E80 max)	**Standard SSD**
Tiered ~20K (P30 5K, P40 7.5K, P60 16K)	**Premium SSD**
Exact IOPS low cost, size decoupled	**Premium SSD v2**
≥20K IOPS or sub-ms latency	**Ultra Disk**

D2 SQL data protection

- Hide sensitive values at query time → **DDM**
- Encrypt whole DB at rest → **TDE**
- Encrypt column from DBAs → **Always Encrypted**
- Restrict who can query table → **SQL roles**
- Audit who queried what / wire → **Auditing / TLS**

DDM masks results (XXX-XX-1234); RBAC/SQL roles control access, not visibility. TDE is default ON for Azure SQL.

D1 boundary + identity location

- Non-Azure infra into ARM → **Arc**
- Cross-tenant RBAC / MSP SSO → **Lighthouse**
- Cross-tenant data snapshot/in-place → **Data Share**
- Azure compute / Arc-enabled / other cloud → **MI / Arc MI / SP**

Real person = user + CA/MFA/PIM. On-prem no Arc or other cloud = SP (cert > secret). Never PATs for app auth.

D3 Key Vault DR trio

- TDE protector key type/sizes → **RSA 2048/3072/4096**
- Never for TDE protector → **AES**
- Auto-failover crypto ops → **Work**
- Auto-failover writes → **Blocked**
- Customer backup/restore → **Same geography***

Failover = MS-managed paired region, read-only. Get/Wrap/Unwrap/Encrypt/Decrypt work; Set/Create/Delete blocked. *Brazil South ↔ South Central US is the cross-geo backup/restore exception.

D3 DB HA / DR tier traps

Requirement	Pick
MySQL/PostgreSQL zone outage	**Zone-redundant HA**
Region fails, min cost	**Geo-redundant backup**
Lower RTO/RPO regional DR	**Cross-region read replica**
SQL DB DTU zone redundant	**Premium only**
SQL DB vCore zone redundant	**GP / BC / Hyperscale**

Exam trap: cheapest region DR for Flexible Server = geo-backup restore, not read replica. BC has sync replicas / zero data loss for AZ failover.

D1 app registration + Graph

Blade / suffix	Meaning
App roles	Roles I offer to callers (API/server)
API permissions	Perms I want to call (client)
Expose an API	Delegated scopes + authorized clients
Token config	Optional claims
Enterprise app assignment required	Yes + assign SPs to app role
No suffix / .Shared / .All	Own data / shared with user / tenant-wide

Application permissions require admin consent only; no user consent. Use assignment required when only specific service principals may call.

D2 SAS + Files auth

- Shared key → **Avoid**
- Account SAS / Service SAS → **Key-signed**
- Revocable Service SAS → **Stored access policy**
- Max security, no key, time-limited → **User delegation SAS**
- Azure Files SMB modern / legacy → **Entra Kerberos / AD DS**

Signer RBAC: Blob Data Reader = read SAS; Contributor = write SAS; Owner = ACL. Entra Domain Services also legacy-ish for SMB.

D1 Log Analytics cost levers

Requirement	Pick
Hot query default 30d→2y (~$2.30/GB)	**Analytics/Interactive**
Must ingest all + minimize cost (~75% less)	**Basic Logs**
Even cheaper / long retention up to 12y	**Auxiliary / Archive**
Known daily volume discount	**Commitment tiers**
Drop/edit before ingest	**DCR transform**
Query Archive / hard drop ceiling	**Search job or Restore / cap**

Basic Logs: cheaper ingest, pay-per-query; Simple Log Alerts supported (≥1-min freq; 100/5K limit for 1-min rules). Archive ≈$0.02/GB-mo; query via Search jobs or Restore.

D4 SQL migration + moves

- AG-source → SQL MI bridge → **MI Link**
- AG-source / cross-OS seconds cutover → **Distributed AG**
- Single SQL online to SQL DB/MI/VM → **DMS online**
- Post-migration SQL MI cross-region DR → **Auto-failover group**
- Small DBs / same-region / cross-region / DR only → **Backup URL / move / RM / ASR**

MI Link uses Distributed AG under the hood. Auto-failover group: Microsoft-managed, planned = no data loss, unplanned may lose recent writes.

D4 AKS scaling

- More pods in existing deployment → **HPA**
- More nodes in a node pool → **Cluster Autoscaler**
- Bursty Linux stateless pods → **Virtual Nodes**
- Event-driven scale-to-zero → **KEDA**

HPA scales pods; Cluster Autoscaler scales nodes; Virtual Nodes burst to ACI (Linux only).

D4 App Service plan vs ASE

- Normal web app hosting / shared platform → **App Service plan**
- Need private inbound only → **Private Endpoint**
- Single-tenant dedicated VNet env → **ASE v3**
- >30 instances / high scale / compliance isolation → **ASE v3**

Plan = compute profile for apps; apps in same plan share workers. ASE = isolated App Service stamp in your subnet; uses Isolated v2 plans.

Good luck!